



Advanced Time Series Analysis and Forecasting

Seminar 10: Long memory and rough volatility



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Acronyms used in this material (1/2)

- ▣ **ACF**: Autocorrelation Function
- ▣ **AI**: Artificial Intelligence
- ▣ **AR**: AutoRegressive (model); in event studies: Abnormal Return
- ▣ **ARCH**: AutoRegressive Conditional Heteroskedasticity
- ▣ **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- ▣ **ARMA**: AutoRegressive Moving Average (model)
- ▣ **BET**: Bucharest Exchange Trading (index)
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **BUX**: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- ▣ **CAC**: Cotation Assistée en Continu (CAC 40 index, Paris) — the CAC 40 index of the Paris bourse
- ▣ **CEE**: Central and Eastern Europe
- ▣ **DM**: Diebold–Mariano (test)
- ▣ **ELW**: Exact Local Whittle (estimator)
- ▣ **EODHD**: EOD Historical Data (market-data provider)
- ▣ **FCVAR**: Fractionally Cointegrated VAR
- ▣ **FFT**: Fast Fourier Transform
- ▣ **FIGARCH**: Fractionally Integrated GARCH

Acronyms used in this material (2/2)

- ▣ **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GPH**: Geweke–Porter–Hudak (estimator)
- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **HAR**: Heterogeneous AutoRegressive (model)
- ▣ **LW**: Local Whittle (estimator)
- ▣ **MSE**: Mean Squared Error
- ▣ **OLS**: Ordinary Least Squares
- ▣ **PX**: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- ▣ **QLIKE**: Quasi-LIKElihood loss
- ▣ **RFSV**: Rough Fractional Stochastic Volatility (model)
- ▣ **RV**: Realised Variance
- ▣ **SD**: Standard Deviation
- ▣ **SE**: Standard Error
- ▣ **VECM**: Vector Error Correction Model
- ▣ **WIG20**: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Today's question and seminar structure (1/4)

- The question of the seminar
 - ▶ **how do we measure, test and use long memory and roughness in volatility and inflation?**
- Three types of exercises
 - ▶ **A**: calculations and derivations on paper
 - ▶ **B**: real data, solved in the notebook
 - ▶ **C**: open questions, for example correcting an answer given by an AI tool
- Each exercise is marked
 - ▶ **[Solved]**: the full solution is presented in the seminar, in the slides and in the notebook; it is the model to follow
 - ▶ **[Proposed]**: you solve it yourselves, following the model; the solution is discussed in the seminar

Today's question and seminar structure (2/4)

- Part A: derivations and computations on paper
 - ▶ A1: ARFIMA autocorrelations and the mean
 - ▶ A2: fractional differencing weights
 - ▶ A3: standard errors of GPH and local Whittle
 - ▶ A4: bias and the optimal bandwidth
 - ▶ A5: fractional Gaussian noise and scaling
 - ▶ A6: the RFSV kernel and the variance correction
 - ▶ A7: circulant embedding by hand
 - ▶ A8: switching that mimics long memory

Today's question and seminar structure (3/4)

- ▣ Part B: real data, inference and interpretation
 - ▶ B1: memory of FTSE 100 realised variance
 - ▶ B2: BET and EUR/RON with the Qu test
 - ▶ B3: roughness of the Euro Stoxx 50
 - ▶ B4: Bitcoin roughness by sampling interval
 - ▶ B5: forecasting FTSE 100 realised variance
 - ▶ B6: forecasting CAC 40 realised variance
- ▣ Part C: open questions and AI critique
 - ▶ C1: co-memory of DAX and CAC 40 volatility
 - ▶ C2: audit an AI answer

Today's question and seminar structure (4/4)

- ▣ Outputs to produce
 - ▶ the answers to the A exercises
 - ▶ one figure and one table for each B exercise
 - ▶ the analysis requested in C1 and the verdicts on the AI answer in C2, each with one line of justification
 - ▶ one sentence of interpretation for each exercise
- ▣ Notebook for today: [open the seminar notebook in Google Colab](#)
 - ▶ each task names its notebook section
- ▣ Seminar: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	ARFIMA autocorrelations and the mean; fractional weights	Solved, Proposed	A1
A3, A4	standard errors of GPH and LW; bias and bandwidth	Solved, Proposed	A3
A5, A6	fractional Gaussian noise; the RFSV kernel	Solved, Proposed	A5
A7, A8	circulant embedding by hand; switching that mimics long memory	Solved, Proposed	A7, A1
B1, B2	memory of FTSE 100 log RV; BET and EUR/RON with the Qu test	Solved, Proposed	B1
B3, B4	roughness of the Euro Stoxx 50; Bitcoin by sampling interval	Solved, Proposed	B3
B5, B6	forecasting RV of the FTSE 100 and the CAC 40	Solved, Proposed	B5
C1, C2	fractional cointegration of two volatilities; what is wrong in an AI answer?	Proposed	B1, A1–A8

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
RV of FTSE 100, Euro Stoxx 50, DAX, CAC 40	Heber et al. (2009)	daily, 5-minute returns	2000–2022
RV of Bitcoin	Binance one-minute prices	daily, 1–60 minutes	2018–2026
BET	EODHD	daily	2000–2026
EUR/RON	BNR reference rate	daily	2005–2026

- ▣ RV in $\%^2$, returns in %
- ▣ Functions in the notebook
 - ▶ `gph`, `local_whittle`, `elw`, `qu_stat`, `gjr_scaling`, `h_with_noise`, `rfsv_kernel`, `forecast_rv`, `fcvar_fit`
- ▣ No account or key is needed

Prerequisites for Today (1/24): autocorrelation and memory

- Autocovariance and autocorrelation of a stationary series X_t at lag k

$$\gamma(k) = \text{Cov}(X_t, X_{t+k}), \quad \rho(k) = \frac{\gamma(k)}{\gamma(0)} \in [-1, 1]$$

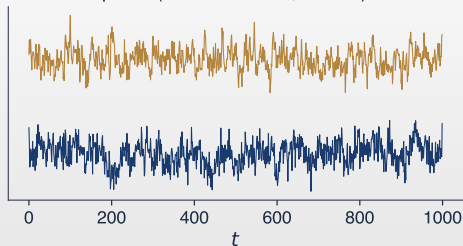
- $\gamma(0) = \text{Var}X_t$; stationary: mean and $\gamma(k)$ do not depend on t
- Short memory:** autocorrelations decay exponentially and are summable
 - AR(1) $X_t = \phi X_{t-1} + \varepsilon_t$, $|\phi| < 1$: $\rho(k) = \phi^k$, so $\sum_k |\rho(k)| < \infty$
 - ε_t : white noise (uncorrelated, mean 0, variance σ^2)
- Long memory:** hyperbolic decay, not summable, a spectral pole at frequency zero

$$\rho(k) \sim c k^{2d-1} \ (k \rightarrow \infty), \quad f(\lambda) \sim G \lambda^{-2d} \ (\lambda \rightarrow 0), \quad 0 < d < 1/2$$

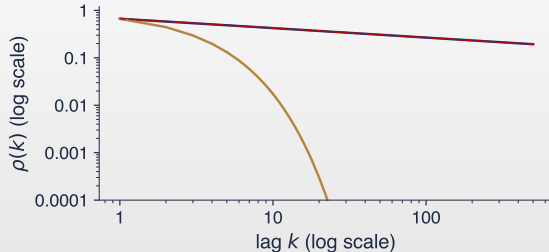
- d : memory parameter; $c, G > 0$: constants; $f(\lambda)$: spectral density at frequency λ (defined below)
- $a_k \sim b_k$: the ratio $a_k/b_k \rightarrow 1$; Hurst exponent $H = d + 1/2$

Prerequisites for Today (2/24): hyperbolic against exponential decay

Simulated paths (same variance, shifted)



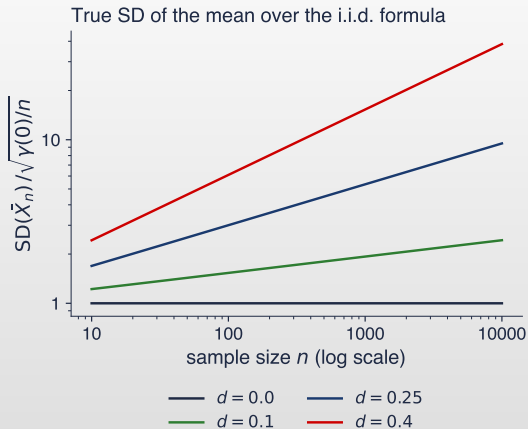
Autocorrelations



— $\text{AR}(1), \phi = 0.67$
 — $\text{ARFIMA}(0, 0.4, 0)$
 --- $\frac{\Gamma(1-d)}{\Gamma(d)} k^{2d-1}$

- $\text{ARFIMA}(0, 0.4, 0)$ (defined below) and an $\text{AR}(1)$ with the same $\rho(1)$, $\phi = 0.67$; both simulated, $n = 1000$
- At lag 100: $\rho = 0.27$ for ARFIMA, practically 0 for the $\text{AR}(1)$; on log scales the hyperbolic ACF is a straight line of slope $2d - 1$
- The ARFIMA path wanders in long swings around its mean

Prerequisites for Today (3/24): the variance of the sample mean



- For any stationary series

$$\text{Var}(\bar{X}_n) = \frac{1}{n} \sum_{|k| < n} \left(1 - \frac{|k|}{n}\right) \gamma(k)$$

- ▶ $\bar{X}_n = \frac{1}{n} \sum_{t=1}^n X_t$; n : sample size
- ▶ i.i.d.: only $k = 0$ remains, $\gamma(0)/n$
- Long memory: of order n^{2d-1} , not n^{-1}
 - ▶ the ratio in the chart grows like n^d
 - ▶ $n = 1000$: 1.9, 5.3, 15.3 for $d = 0.1, 0.25, 0.4$
 - ▶ the i.i.d. standard error is that many times too small

Prerequisites for Today (4/24): the lag operator and fractional differencing

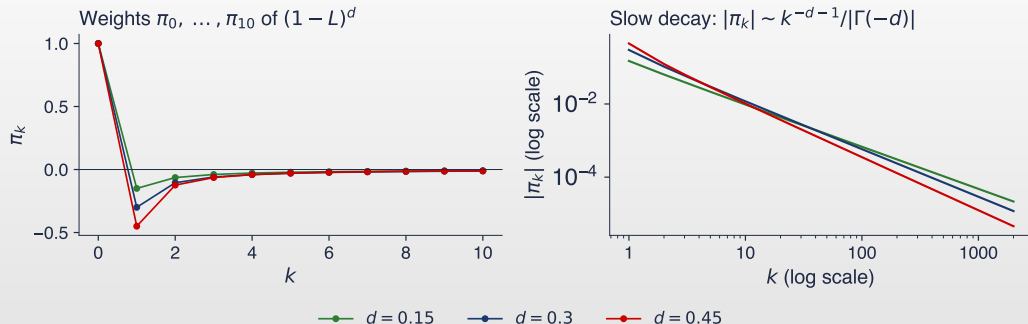
□ **Lag operator** L : $LX_t = X_{t-1}$, $L^k X_t = X_{t-k}$; first difference $(1 - L)X_t = X_t - X_{t-1}$

□ **Fractional difference**: the binomial series of $(1 - L)^d$ for a real d

$$(1 - L)^d X_t = \sum_{k=0}^{\infty} \pi_k X_{t-k}, \quad \pi_0 = 1, \quad \pi_k = \pi_{k-1} \frac{k-1-d}{k}$$

- ▶ π_k : weight of X_{t-k} ; $d = 1$: $\pi_1 = -1$, $\pi_k = 0$ for $k \geq 2$ (the usual difference)
- ▶ $0 < d < 1$: all past values enter, with weights $\pi_k \sim k^{-d-1}/\Gamma(-d)$
- ▶ $\Gamma(z) = \int_0^{\infty} u^{z-1} e^{-u} du$: gamma function, $\Gamma(n) = (n-1)!$; $\Gamma(-d) < 0$ for $0 < d < 1$
- ▶ $\sum_k \pi_k = (1-1)^d = 0$: a constant level is removed only by the infinite filter

Prerequisites for Today (5/24): fractional weights



- Left: $\pi_1 = -d$, then small negative weights that shrink slowly
- Right: on log scales $|\pi_k|$ is a straight line of slope $-(d+1)$: a hyperbolic, not exponential, decay
- A filter truncated after K lags leaves out the tail of the weights

Prerequisites for Today (6/24): the ARFIMA model

- **ARFIMA**(0, d , 0) [Hosking \(1981\)](#): white noise after fractional differencing

$$(1 - L)^d X_t = \varepsilon_t, \quad -\frac{1}{2} < d < \frac{1}{2} \text{ (stationary and invertible)}$$

- ▶ $d > 0$: long memory; $d < 0$: antipersistence ($\sum_k \rho(k) = 0$); $d \geq 1/2$: nonstationary

- Autocorrelations: a recursion and its limit

$$\rho(0) = 1, \quad \rho(k) = \rho(k-1) \frac{k-1+d}{k-d}, \quad \rho(k) \sim \frac{\Gamma(1-d)}{\Gamma(d)} k^{2d-1}$$

- **ARFIMA**(1, d , 0): $(1 - \phi L)(1 - L)^d X_t = \varepsilon_t$

- ▶ ϕ : AR coefficient, $|\phi| < 1$; it changes the short lags, d the long ones
- ▶ **ARFIMA**(p , d , q): an **ARMA**(p , q) for $(1 - L)^d X_t$

Prerequisites for Today (7/24): the spectral density and the periodogram

- **Spectral density:** the variance of X_t decomposed by frequency

$$f(\lambda) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \gamma(k) e^{-i\lambda k}, \quad \gamma(0) = \int_{-\pi}^{\pi} f(\lambda) d\lambda$$

- ▶ $\lambda \in [-\pi, \pi]$: angular frequency, a cycle of $2\pi/\lambda$ periods; low λ : long cycles
- ▶ ARFIMA(0, d , 0): $f(\lambda) = \frac{\sigma^2}{2\pi} (4 \sin^2(\lambda/2))^{-d} \sim \frac{\sigma^2}{2\pi} \lambda^{-2d}$

- **Periodogram** at the Fourier frequencies: the sample counterpart of f

$$I(\lambda_j) = \frac{1}{2\pi n} \left| \sum_{t=1}^n X_t e^{-i\lambda_j t} \right|^2, \quad \lambda_j = \frac{2\pi j}{n}, \quad j = 1, \dots, \lfloor n/2 \rfloor$$

- ▶ $I(\lambda_j)/f(\lambda_j)$ is approximately exponential with mean 1, independent across j : very noisy
- ▶ bandwidth m : the number of lowest frequencies $\lambda_1, \dots, \lambda_m$ used to estimate d ; $m \rightarrow \infty$, $m/n \rightarrow 0$

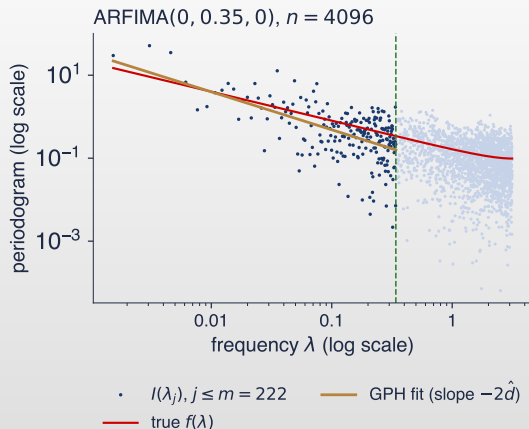
Prerequisites for Today (8/24): the GPH estimator

- **GPH Geweke and Porter-Hudak (1983):** take logs of $I(\lambda_j) \approx f(\lambda_j) \xi_j$ near zero and run OLS

$$\log I(\lambda_j) = a + d \left[-\log \left(4 \sin^2(\lambda_j/2) \right) \right] + u_j, \quad j = 1, \dots, m$$

- ▶ ξ_j : exponential error with mean 1; $u_j = \log \xi_j - \mathbb{E} \log \xi_j$; a : intercept
- ▶ $\hat{d}$: the OLS slope; regressor $\approx -2 \log \lambda_j$, so the log-log slope is $-2d$
- Asymptotic standard error
 - ▶ $\text{Var} u_j = \pi^2/6$ and the regressor has variance ≈ 4 : $\text{SE}(\hat{d}) = \pi/\sqrt{24m}$
 - ▶ $\sqrt{m}(\hat{d} - d) \rightarrow N(0, \pi^2/24)$; 95% interval $\hat{d} \pm 1.96 \text{ SE}$
- Precision grows only with $\sqrt{m}$, not with $\sqrt{n}$: estimating d needs long samples

Prerequisites for Today (9/24): the log-periodogram near zero



- Simulated ARFIMA(0, 0.35, 0), $n = 4096$
- The $m = n^{0.65} = 222$ lowest ordinates (dark) lie around a line
 - ▶ GPH: $\hat{d} = 0.46$; local Whittle: 0.44
 - ▶ both within two SEs of $d = 0.35$
- The scatter around the line is the exponential noise ξ_j : it does not shrink with n
- Far from zero (pale points) the short-run dynamics bend f : they must stay out of the regression

Prerequisites for Today (10/24): the local Whittle estimator

- Local Whittle Robinson (1995b): Gaussian likelihood of $f(\lambda_j) = G\lambda_j^{-2d}$ on the m lowest frequencies

$$\hat{d} = \arg \min_d R(d), \quad R(d) = \log \hat{G}(d) - \frac{2d}{m} \sum_{j=1}^m \log \lambda_j, \quad \hat{G}(d) = \frac{1}{m} \sum_{j=1}^m \lambda_j^{2d} I(\lambda_j)$$

- $\hat{G}(d)$: the level G that fits best for a given d ; $R(d)$: the profiled objective (a QLIKE-type fit)

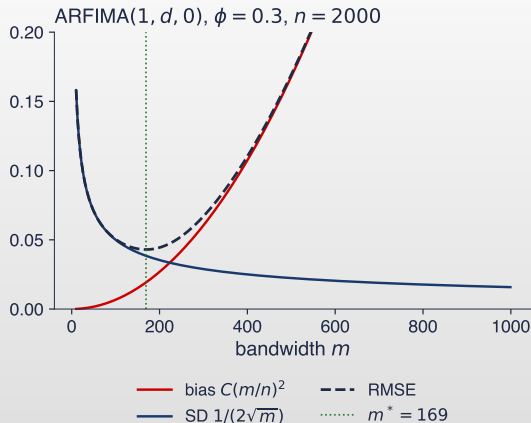
- Inference

- $\sqrt{m}(\hat{d} - d) \rightarrow N(0, 1/4)$: $SE = 1/(2\sqrt{m})$, smaller than GPH at the same m
 - relative efficiency: the ratio of the two asymptotic variances, $(\pi^2/24)/(1/4)$
 - consistent for $-1/2 < d \leq 1$; for $d > 1$ it converges to 1

- Exact local Whittle (ELW) Shimotsu and Phillips (2005), Shimotsu (2010)

- replaces $\lambda_j^{2d} I(\lambda_j)$ by the periodogram of $(1 - L)^d(X_t - \hat{\mu})$, $\hat{\mu}$: estimated mean
 - $N(d, 1/(4m))$ for any d in the search range, also $d \geq 1/2$ (nonstationary)

Prerequisites for Today (11/24): bias and the choice of bandwidth



- Short-run dynamics bias $\hat{d}$; for ARFIMA(1, d , 0)

$$\mathbb{E}\hat{d} - d \approx C \left(\frac{m}{n} \right)^2, \quad C = \frac{2\pi^2}{9} \cdot \frac{2\phi}{(1-\phi)^2}$$

- $\phi > 0$: positive bias, growing with m

- MSE = bias² + $\frac{1}{4m}$; minimised at

$$m^* = \left(\frac{n^4}{16C^2} \right)^{1/5}$$

- MSE: mean squared error; m^* grows like $n^{4/5}$
- chart: $\phi = 0.3$, $n = 2000$: $C = 2.69$,
 $m^* = 169 = n^{0.68}$

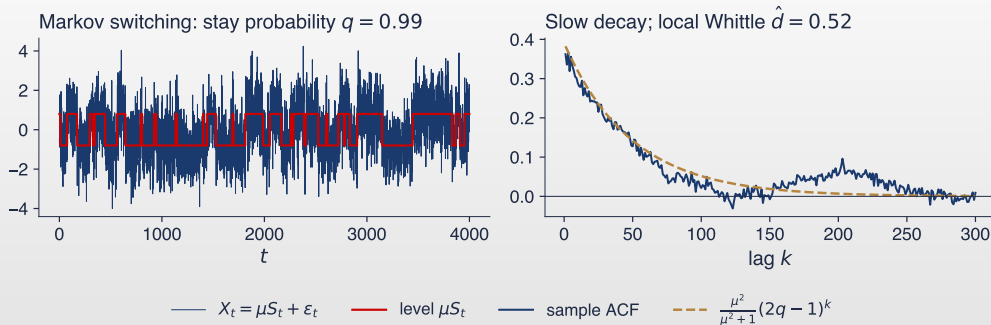
Prerequisites for Today (12/24): level shifts that mimic long memory

- **Two-state Markov chain** $S_t \in \{-1, 1\}$: it stays with probability q , switches with $1 - q$

$$X_t = \mu S_t + \varepsilon_t, \quad \mathbb{E}(S_t S_{t+k}) = (2q - 1)^k$$

- ▶ μ : size of the level; ε_t : i.i.d. $N(0, 1)$, independent of S ; expected duration of a regime $1/(1 - q)$
- ▶ the ACF decays exponentially: short memory in the strict sense
- **Half-life** of a geometric decay $\rho(k) = \rho(0) r^k$
 - ▶ $k_{1/2} = \ln 0.5 / \ln r$: the lag at which the autocorrelation has halved
- Rare shifts, structural breaks and slowly varying trends put power at the lowest frequencies [Diebold and Inoue \(2001\)](#)
 - ▶ $\hat{d}$ is then large although the process has short memory: **spurious long memory**

Prerequisites for Today (13/24): a simulated switching series



- $q = 0.99$ (regimes last 100 periods on average), $\mu = 0.8$, $n = 4000$
- Local Whittle with $m = n^{0.65}$ gives $\hat{d} = 0.52$, a value typical of a long-memory series
- Over a few hundred lags the slow geometric decay cannot be told apart from a hyperbolic one

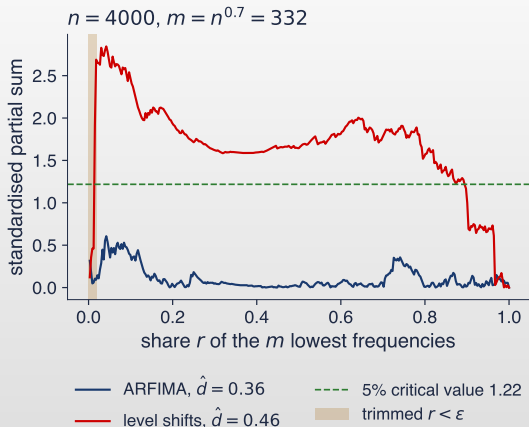
Prerequisites for Today (14/24): the Qu test

- **Qu test** [Qu \(2011\)](#): H_0 stationary long memory, $f(\lambda) = G\lambda^{-2d}$ on the m lowest frequencies

$$W = \sup_{r \in [\varepsilon, 1]} \left(\sum_{j=1}^m \nu_j^2 \right)^{-1/2} \left| \sum_{j \leq mr} \nu_j \left(\frac{\lambda_j^{2\hat{d}} I(\lambda_j)}{\hat{G}} - 1 \right) \right|$$

- ▶ $\hat{d}$, $\hat{G} = \hat{G}(\hat{d})$: local Whittle estimates; $\nu_j = \log \lambda_j - \frac{1}{m} \sum_i \log \lambda_i$
 - ▶ the bracket is the score of local Whittle at frequency j ; its full sum is zero at $\hat{d}$
 - ▶ r : share of the m frequencies summed; ε : trimming, the first εm partial sums are skipped
- Reading
- ▶ under H_0 the partial sums stay small at every r ; level shifts make the lowest ordinates too large: a big W
 - ▶ reject for W above the critical value; with $m = n^{0.7}$, $\varepsilon = 0.02$ (simulated): 1.09 (10%), 1.22 (5%), 1.56 (1%)

Prerequisites for Today (15/24): the Qu partial sums



- Two simulated series, $n = 4000$
- ARFIMA(0, 0.35, 0): the curve stays low
 - ▶ $W = 0.61$: do not reject
- Markov level shifts ($q = 0.997, \mu = 0.8$)
 - ▶ $\hat{d}$ looks like long memory
 - ▶ but the curve jumps at small r : $W = 2.85$, reject at 1%
- A rejection says only that stationary long memory does not fit: breaks, shifts or trends are alternatives

Prerequisites for Today (16/24): fractional cointegration

- **Fractional cointegration:** p series, each with memory d , whose combinations $\beta'X_t$ have memory $d - b$, $b > 0$
- **FCVAR Johansen and Nielsen (2012):** a VECM written in fractional differences

$$\Delta^d X_t = \alpha \beta' L_b \Delta^{d-b} X_t + \sum_{i=1}^k \Gamma_i \Delta^d L_b^i X_t + \varepsilon_t, \quad \Delta^d = (1 - L)^d, \quad L_b = 1 - \Delta^b$$

- ▶ β ($p \times r$): cointegrating vectors; α ($p \times r$): adjustment speeds; r : cointegration rank
- ▶ L_b : fractional lag operator (only past values); k : number of lagged short-run terms, Γ_i their matrices; $b = d = 1$: the usual VECM
- Estimation and rank tests
 - ▶ for fixed (d, b) : Johansen reduced-rank regression; profile likelihood: maximise over a grid of (d, b)
 - ▶ trace test of rank r against p : $LR = 2(\ell_p - \ell_r)$, ℓ : maximised log likelihoods; $\chi^2_{(p-r)^2}$ when $b < 1/2$

Prerequisites for Today (17/24): fractional Brownian motion

- **Fractional Brownian motion** (fBm) W_t^H [Mandelbrot and Van Ness \(1968\)](#): Gaussian, $W_0^H = 0$, mean 0, covariance

$$\mathbb{E}(W_t^H W_s^H) = \frac{1}{2} (t^{2H} + s^{2H} - |t - s|^{2H}), \quad \mathbb{E}|W_{t+\Delta}^H - W_t^H|^2 = \Delta^{2H}$$

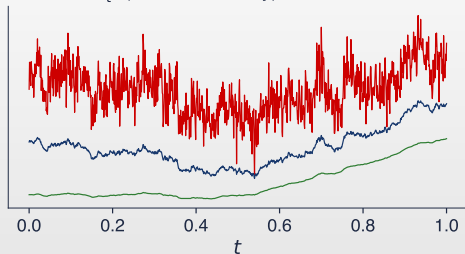
- ▶ $H \in (0, 1)$: Hurst exponent; $H = 1/2$: Brownian motion, independent increments
- ▶ $H < 1/2$: rough (irregular) paths, negatively correlated increments; $H > 1/2$: smooth paths, persistent increments

- **Fractional Gaussian noise** (fGn): the unit increments $W_{t+1}^H - W_t^H$

$$\gamma_k = \rho(k) = \frac{1}{2} (|k+1|^{2H} - 2|k|^{2H} + |k-1|^{2H})$$

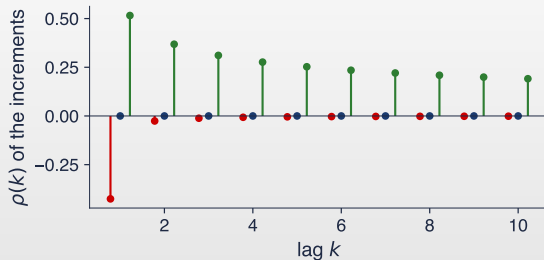
- ▶ $\rho(1) = 2^{2H-1} - 1$; for $H > 1/2$, $\rho(k) \sim H(2H-1)k^{2H-2}$: long memory with $d = H - 1/2$

Prerequisites for Today (18/24): rough and smooth paths

Paths of W_t^H (shifted vertically)

— $H = 0.1$ — $H = 0.5$ — $H = 0.8$

Fractional Gaussian noise



- Left: fBm paths from the same random numbers for $H = 0.1, 0.5, 0.8$
- Right: $H = 0.1$ gives $\rho(1) \approx -0.43$ and almost nothing beyond: the path reverts at once
- $H = 0.8$: positive, slowly decaying autocorrelations, the long memory of the previous frames

Prerequisites for Today (19/24): circulant embedding

- **Goal:** simulate n points of a stationary Gaussian series with autocovariances $\gamma_0, \dots, \gamma_{n-1}$ exactly
 - ▶ Cholesky factorisation of the $n \times n$ covariance costs $O(n^3)$
- **Davies–Harte** [Davies and Harte \(1987\)](#): embed the covariance in a circulant matrix of size $M = 2(n - 1)$

$$c = (\gamma_0, \gamma_1, \dots, \gamma_{n-1}, \gamma_{n-2}, \dots, \gamma_1), \quad \lambda_k = \sum_{j=0}^{M-1} c_j e^{-2\pi i j k / M}$$

- ▶ circulant: each row is the previous one shifted by one place; c : its first row
- ▶ λ_k : its eigenvalues, the FFT (fast Fourier transform) of c , real because c is symmetric
- ▶ if all $\lambda_k \geq 0$: $Y = \text{FFT}(\sqrt{\lambda/M} Z)$, Z complex Gaussian; the first n values have the target covariance
- ▶ cost $O(n \log n)$; for fGn all $\lambda_k \geq 0$ for every H

Prerequisites for Today (20/24): realised variance and moment scaling

- ▣ **Realised variance** of day t from N_t intraday returns $r_{t,i}$ (e.g. 5-minute)

$$RV_t = \sum_{i=1}^{N_t} r_{t,i}^2, \quad \sigma_t = \sqrt{RV_t}$$

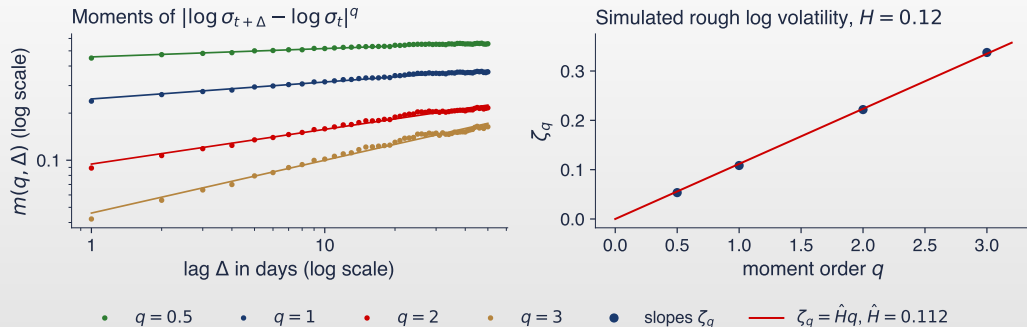
- ▶ a noisy measure of the daily variance; fewer returns per day: a larger measurement error

- ▣ **Moment scaling** Gatheral, Jaisson and Rosenbaum (2018): for $\log \sigma_t = \nu W_t^H$ the moments of the increments are power laws

$$m(q, \Delta) = \frac{1}{N} \sum_t |\log \sigma_{t+\Delta} - \log \sigma_t|^q \propto \Delta^{\zeta_q}, \quad \zeta_q = qH$$

- ▶ q : moment order; Δ : lag in days; N : number of pairs; ν : volatility of volatility (vol-of-vol)
- ▶ ζ_q : slope of $\log m(q, \Delta)$ on $\log \Delta$; then $\hat{H} = \sum_q q \zeta_q / \sum_q q^2$ (regression through the origin)
- ▶ i.i.d. measurement error of variance s^2 adds an intercept: $m(2, \Delta) = \nu^2 \Delta^{2H} + 2s^2$

Prerequisites for Today (21/24): the scaling regression



- Simulated daily log volatility $0.3 W_t^H$ with $H = 0.12$, 4000 days, no measurement error
- Left: straight lines on log scales, one per q ; right: the slopes lie on a line through the origin, $\hat{H} = 0.112$
- Departures from linearity in q (multifractality, heavy-tailed increments) show up at large q

Prerequisites for Today (22/24): HAR, FIGARCH and lognormal forecasts

- **HAR Corsi (2009)**: daily, weekly and monthly averages of the past, $y_t = \log RV_t$

$$y_{t+h} = \beta_0 + \beta_d y_t + \beta_w \bar{y}_t^{(5)} + \beta_m \bar{y}_t^{(22)} + u_{t+h}, \quad \bar{y}_t^{(K)} = \frac{1}{K} \sum_{j=0}^{K-1} y_{t-j}$$

- ▶ h : horizon; OLS; a short-memory AR(22) with restricted weights that mimics long memory

- **FIGARCH Baillie, Bollerslev and Mikkelsen (1996)**: the GARCH variance with a fractional difference of the squared shocks e_t^2

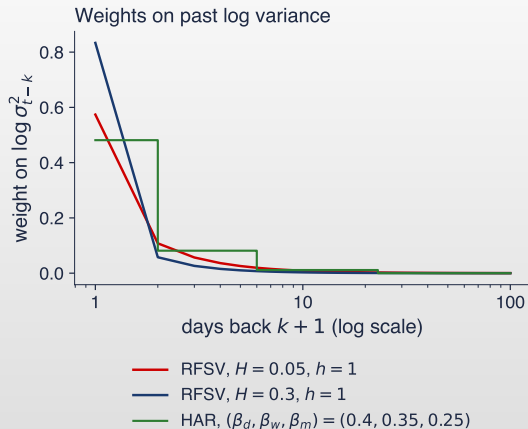
$$\sigma_t^2 = \omega + [1 - (1 - \beta L)^{-1} (1 - \varphi L)(1 - L)^d] e_t^2$$

- ▶ $\omega > 0$, β , φ : GARCH-type coefficients; the bracket is an ARCH(∞) filter whose weights sum to one

- **Lognormal correction**: models forecast $\log RV$, the target is RV

- ▶ if $\log RV_{t+h} \sim N(\mu, s^2)$: $\mathbb{E}RV_{t+h} = e^{\mu+s^2/2}$; e^μ alone is the median, too low

Prerequisites for Today (23/24): the RFSV predictor



- RFSV: $\log \sigma_t = \nu W_t^H$; the forecast is a power-kernel average of the past

$$\widehat{\log \sigma_{t+h}^2} = \sum_k w_k \log \sigma_{t-k}^2$$

- ▶ $w_k \propto \int_k^{k+1} \frac{du}{(u+h)u^{H+1/2}}$, normalised to sum to 1; h : horizon

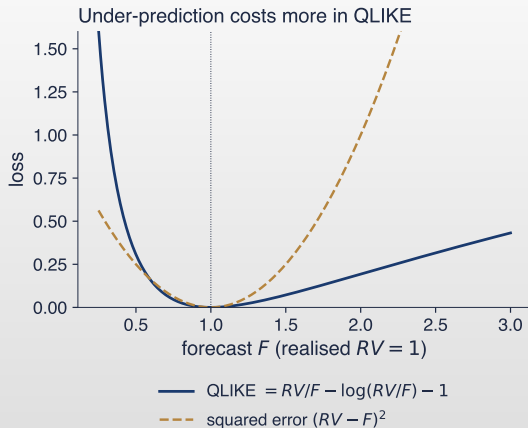
- Variance forecast

$$\exp(\widehat{\log \sigma_{t+h}^2} + 2c\nu^2 h^{2H})$$

- ▶ $c = \frac{\Gamma(3/2-H)}{\Gamma(H+1/2)\Gamma(2-2H)}$; $4c\nu^2 h^{2H}$: the forecast-error variance of $\log \sigma^2$, halved as in the lognormal correction

- Chart: the kernel puts most weight on the last day and a long thin tail behind; HAR uses three flat steps

Prerequisites for Today (24/24): QLIKE and the Diebold–Mariano test



- **QLIKE** [Patton \(2011\)](#) of a variance forecast F

$$\text{QLIKE} = \frac{RV}{F} - \log \frac{RV}{F} - 1 \geq 0$$

- ▶ 0 only if $F = RV$; ranks forecasts correctly although RV is a noisy proxy
- ▶ penalises under-prediction more (chart)

- **DM test** ([Chapter 1](#)): δ_t = loss of model 1 minus loss of model 2

$$DM = \frac{\bar{\delta}}{\sqrt{\hat{\Omega}/T}} \rightarrow N(0, 1)$$

- ▶ $\bar{\delta}$: mean over T forecasts; $\hat{\Omega}$: Newey–West variance $\hat{\gamma}_0 + 2 \sum_{k=1}^L (1 - \frac{k}{L+1}) \hat{\gamma}_k$ of δ_t
- ▶ $DM < -1.96$: model 1 significantly better at 5%

A1: ARFIMA autocorrelations and the mean [Solved]

Task

- ▣ ARFIMA(0, 0.3, 0).
- ▣ 1. Show that $\rho(1) = d/(1 - d)$ and compute $\rho(1)$, $\rho(10)$, $\rho(100)$ with the recursion.
- ▣ 2. Compare $\rho(100)$ with $\frac{\Gamma(1-d)}{\Gamma(d)} 100^{2d-1}$.
- ▣ 3. For $n = 1000$, compute the ratio of the true SD of $\bar{X}$ to the i.i.d. formula $\sqrt{\gamma(0)/n}$ (notebook A1).
- ▣ Report: three numbers; two numbers; one number and one sentence.

Solution

- ▣ 1. $\rho(1) = \rho(0) \cdot \frac{0+d}{1-d}$; $\rho(1) = 0.429$, $\rho(10) = 0.173$, $\rho(100) = 0.069$
- ▣ 2. Asymptotic formula: 0.069, practically equal already at $k = 100$
- ▣ 3. Ratio 7.55: the i.i.d. standard error is 7.55 times too small
- ▣ Interpretation: with $d = 0.3$, 1000 observations carry the information of far fewer independent ones; HAC with a fixed bandwidth cannot repair this

A2: fractional differencing weights [Proposed]

Solution discussed in class

Task

- ▣ $d = 0.4$. Model: A1.
- ▣ 1. Compute $\pi_1, \dots, \pi_5$ of $(1 - L)^{0.4}$.
- ▣ 2. Compare π_{100} with $100^{-d-1}/\Gamma(-d)$.
- ▣ 3. Compute $\sum_{k=0}^{1000} \pi_k$; what is its limit, and what does the gap mean for a truncated filter?
- ▣ Report: five numbers; two numbers; one number and one sentence.



A3: standard errors of GPH and local Whittle [Solved]

Task

- ▣ $n = 5000$, $m = \lfloor n^{0.65} \rfloor$; a local Whittle estimate $\hat{d} = 0.45$.
- ▣ 1. Compute m and the asymptotic SE of GPH and of local Whittle.
- ▣ 2. Give the 95% interval for d and the relative efficiency of LW against GPH.
- ▣ 3. Which m gives a half-width of 0.05?
- ▣ Report: three numbers; an interval and one number; one number.

Solution

- ▣ 1. $m = 253$; GPH $\pi/\sqrt{24m} = 0.0403$; LW $1/(2\sqrt{m}) = 0.0314$
- ▣ 2. $[0.388, 0.512]$; efficiency $(\pi^2/24)/(1/4) = \pi^2/6 = 1.645$
- ▣ 3. $1.96/(2\sqrt{m}) \leq 0.05 \Rightarrow m \geq 385$
- ▣ Interpretation: the interval includes $1/2$, so stationarity is not established; precision is bought with bandwidth, and bandwidth costs bias (A4)

A4: bias and the optimal bandwidth [Proposed]

Solution discussed in class

Task

- ▣ ARFIMA(1, d , 0) with $\phi = 0.5$, $n = 5000$. Model: [A3](#).
- ▣ 1. Compute the bias constant C .
- ▣ 2. For $m = \lfloor n^{0.8} \rfloor$, compute the bias and the SD of local Whittle.
- ▣ 3. Compute m^* , its exponent $\log m^* / \log n$, and the bias and SD at m^* .
- ▣ Report: one number; two numbers; four numbers and one sentence.

A5: fractional Gaussian noise and scaling [Solved]

Task

- ▣ fGn with $H = 0.1, 0.3$ and 0.7 .
- ▣ 1. Compute $\rho(1) = 2^{2H-1} - 1$ and $\rho(2)$ for each H .
- ▣ 2. Compute $m(2, 22)/m(2, 1) = 22^{2H}$: how much more does log volatility move over a month than over a day?
- ▣ 3. Show that $\zeta_q = qH$ for fBm.
- ▣ Report: six numbers; three numbers; one derivation.

Solution

- ▣ 1. $H = 0.1$: $-0.426, -0.026$; $H = 0.3$: $-0.242, -0.049$; $H = 0.7$: $0.320, 0.189$
- ▣ 2. $1.86, 6.39, 75.75$: with $H = 0.1$ the monthly variance of changes is less than twice the daily one
- ▣ 3. $W_{t+\Delta}^H - W_t^H \sim N(0, \Delta^{2H})$: $\mathbb{E}|\cdot|^q = \Delta^{qH} \mathbb{E}|Z|^q$; take logs
- ▣ Interpretation: rough log volatility moves a lot from day to day but drifts little: strong negative correlation of daily changes

A6: the RFSV kernel and the variance correction [Proposed]

Solution discussed in class

Task

- $H = 0.1$, $\nu = 0.3$, horizon $h = 1$; kernel truncated at 500 days. Model: A5.
- 1. Compute the share of the weight on the last day, the last 5 and the last 22 days (notebook A6).
- 2. Compute $c(H)$ and the correction $2c\nu^2 h^{2H}$ for $h = 1, 5, 22$.
- 3. By how much would ignoring the correction understate the variance forecast at $h = 22$?
- Report: three shares; four numbers; one number.

A7: circulant embedding by hand [Solved]

Task

- ▣ Three points of fGn with $H = 0.3$: $\gamma_0, \gamma_1, \gamma_2$.
- ▣ 1. Compute γ_1 and γ_2 .
- ▣ 2. Write the first row of the 4×4 circulant and its eigenvalues $\lambda_k = \sum_j c_j e^{-2\pi ijk/4}$.
- ▣ 3. Can the Davies–Harte method be used?
- ▣ Report: two numbers; a row and four numbers; one sentence.

Solution

- ▣ 1. $\gamma_1 = \frac{1}{2}(2^{0.6} - 2) = -0.2421$,
 $\gamma_2 = \frac{1}{2}(3^{0.6} - 2 \cdot 2^{0.6} + 1) = -0.0491$
- ▣ 2. $c = (\gamma_0, \gamma_1, \gamma_2, \gamma_1)$; $\lambda_0 = \gamma_0 + 2\gamma_1 + \gamma_2 = 0.4666$,
 $\lambda_1 = \lambda_3 = \gamma_0 - \gamma_2 = 1.0491$, $\lambda_2 = \gamma_0 - 2\gamma_1 + \gamma_2 = 1.4352$
- ▣ 3. Yes: all eigenvalues are positive, so the circulant is a valid covariance and its first 3 coordinates have the fGn covariance
- ▣ Interpretation: the embedding doubles the size but turns an $O(n^3)$ factorisation into two FFTs

A8: switching that mimics long memory [Proposed]

Solution discussed in class

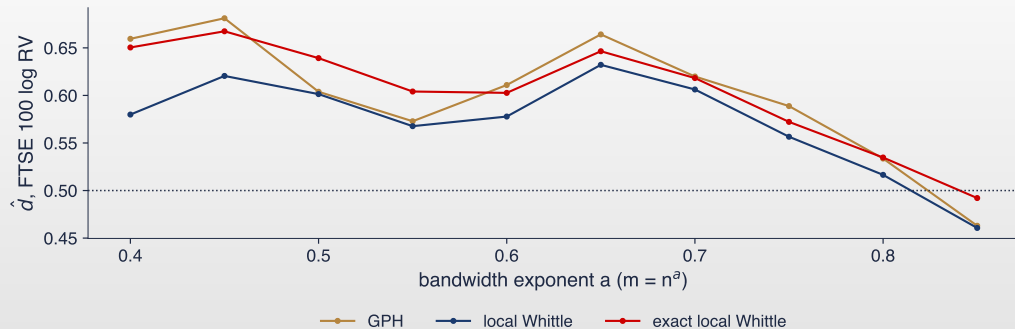
Task

- $X_t = \mu S_t + \varepsilon_t$, $S_t \in \{-1, 1\}$ a Markov chain that stays with probability $q = 0.995$, $\mu = 1$, $\varepsilon_t \sim N(0, 1)$. Model: [A1](#).
- 1. Show that $\rho(k) = \frac{\mu^2}{\mu^2 + 1} (2q - 1)^k$ and compute $\rho(1)$, $\rho(100)$, $\rho(500)$.
- 2. Compute the half-life of the autocorrelation.
- 3. Fit $\log \rho(k)$ on $\log k$ for $k = 5, \dots, 200$ and convert the slope to d .
- Report: three numbers; one number; one number and one sentence.

B1: memory of FTSE 100 realised variance [Solved]

- ▣ **Question:** how large is d for the log realised variance of the FTSE 100, and is it robust to the bandwidth and to level shifts?
- ▣ **Inputs:** Oxford-Man 5-minute RV of the FTSE 100, $n = 5\,586$ days to 25 February 2022
- ▣ **Tasks**
 1. Estimate d of log RV by GPH, local Whittle and exact local Whittle for $m = n^{0.5}, n^{0.6}, n^{0.7}$, with standard errors.
 2. Plot $\hat{d}(m)$ for $m = n^{0.40}, \dots, n^{0.85}$.
 3. Compute the Qu statistic with $m = n^{0.7}$ and $\varepsilon = 0.02$.
 4. Interpretation: is the memory of FTSE 100 volatility genuine, and is it stationary?
- ▣ **Report:** a table of nine estimates with SEs, one chart, one statistic, one sentence
- ▣ **Notebook:** section [B1](#)

B1: solution (1/2) [Solved]



B1: solution (2/2) [Solved]

m	GPH (SE)	LW	ELW	SE (LW, ELW)
74	0.604 (0.075)	0.601	0.639	0.058
177	0.611 (0.048)	0.578	0.603	0.038
419	0.620 (0.031)	0.606	0.618	0.024

- ▣ Qu: $W = 0.33$ against the 10% value 1.09: no evidence of level shifts
- ▣ Interpretation: $\hat{d} \approx 0.6$ at every bandwidth with a flat $\hat{d}(m)$: genuine long memory, at or just beyond the stationarity border; report ELW

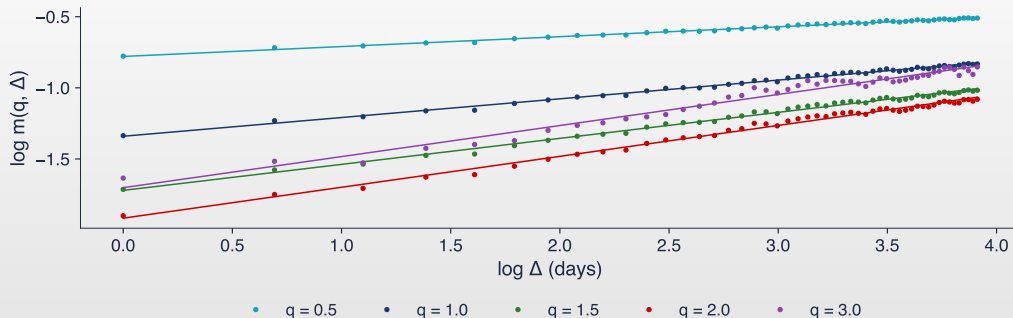
B2: BET and EUR/RON with the Qu test [Proposed]

- **Question:** is the memory of $|r_t|$ on the Bucharest market and for the leu genuine, or does it come from regime changes?
- **Inputs:** BET 2000–2026 and from 2010; EUR/RON (BNR) 2005–2026 and from 2014; model: [B1](#)
- **Tasks**
 1. Estimate d of $|r_t|$ by local Whittle with $m = n^{0.65}$ on the full sample and after the cut date.
 2. Compute the Qu statistic ($m = n^{0.7}$, $\varepsilon = 0.02$) on the same samples.
 3. Interpretation: which of the two markets has genuine long memory in volatility?
- **Report:** four estimates with SEs, four statistics, one sentence
- **Notebook:** section [B2](#)

B3: roughness of the Euro Stoxx 50 [Solved]

- **Question:** does the Gatheral–Jaisson–Rosenbaum scaling hold for the Euro Stoxx 50, and how rough is its volatility?
- **Inputs:** Oxford-Man 5-minute RV, $n = 5\,635$ days; $\sigma_t = \sqrt{RV_t}$; $q = 0.5, 1, 1.5, 2, 3$; $\Delta = 1, \dots, 50$
- **Tasks**
 1. Compute $m(q, \Delta)$ and the slopes ζ_q ; estimate H by regressing ζ_q on q through the origin.
 2. Estimate H with the measurement-error intercept $m(2, \Delta) = \nu^2 \Delta^{2H} + 2s^2$.
 3. Repeat the first estimate on the two halves of the sample.
 4. Interpretation: is ζ_q linear in q , and what does a departure mean?
- **Report:** five slopes, three values of H , ν , one sentence
- **Notebook:** section [B3](#)

B3: solution [Solved]



□ $\zeta_q = 0.069, 0.132, 0.183, 0.217, 0.218$; $\hat{H} = 0.093$, $\hat{\nu} = 0.38$; with the error intercept 0.109; halves 0.091 and 0.095

□ Interpretation: linear up to $q = 2$, flat between $q = 2$ and 3: large moments are driven by a few extreme days (heavy tails of the increments), so H from $q \leq 2$ is the safer reading; roughness is stable over time

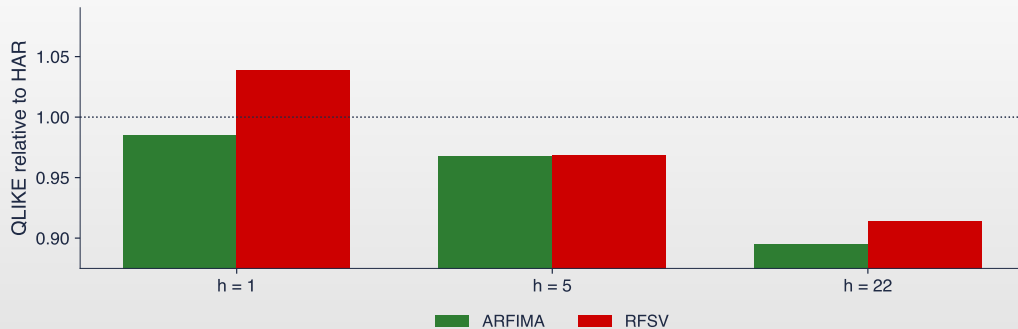
B4: Bitcoin roughness by sampling interval [Proposed]

- **Question:** does the estimate of H for Bitcoin depend on the sampling interval of the realised variance?
- **Inputs:** Binance RV of Bitcoin from 1-, 5-, 15-, 30- and 60-minute returns, 2018–2026; model: [B3](#)
- **Tasks**
 1. Estimate H for each RV by the moment regression and with the error intercept.
 2. Estimate d of log RV by local Whittle with $m = n^{0.65}$.
 3. Interpretation: which direction does a coarser grid push H , and why?
- **Report:** ten values of H , five values of d , one sentence
- **Notebook:** section [B4](#)

B5: forecasting FTSE 100 realised variance [Solved]

- **Question:** does a rough or a long-memory model forecast FTSE 100 realised variance better than HAR?
- **Inputs:** rolling window of 1000 days, re-estimation every 20 days; out of sample 30 December 2003 – 26 January 2022 ($T = 4\,565$); horizons 1, 5, 22 days
- **Tasks**
 1. Forecast RV_{t+h} with HAR, ARFIMA(0, d , 0) and RFSV, with the lognormal corrections.
 2. Compute the average QLIKE of each model and the ratio to HAR.
 3. Compute the DM statistics against HAR.
 4. Interpretation: which model would you use at each horizon?
- **Report:** nine losses, six ratios, six DM statistics, one sentence
- **Notebook:** section [B5](#)

B5: solution (1/2) [Solved]



B5: solution (2/2) [Solved]

h	QLIKE HAR	ARFIMA / HAR	DM	RFSV / HAR	DM
1	0.230	0.985	-0.98	1.038	1.49
5	0.324	0.968	-1.94	0.968	-1.33
22	0.561	0.895	-2.12	0.914	-1.93

- Interpretation: HAR at one day (RFSV is worse); at 22 days both ARFIMA ($p = 0.034$) and RFSV ($p = 0.053$) beat HAR by about a tenth; median rolling $\hat{H} = 0.11$

B6: forecasting CAC 40 realised variance [Proposed]

- **Question:** does the FTSE 100 ranking carry over to the CAC 40?
- **Inputs:** the same design, out of sample 22 December 2003 – 26 January 2022 ($T = 4\,631$); model: [B5](#)
- **Tasks**
 1. Repeat the three forecasts and the QLIKE ratios.
 2. Compute the DM statistics against HAR.
 3. Interpretation: is there a single best model across the two indices?
- **Report:** six ratios, six DM statistics, one sentence
- **Notebook:** section [B6](#)

C1: co-memory of DAX and CAC 40 volatility [Proposed]

- **Question:** do the volatilities of two neighbouring markets share one fractional trend?
- **Inputs:** log 5-minute RV of the DAX and the CAC 40 (Oxford-Man), $n = 5\,600$ common days; FCVAR with $k = 0$; model: [B1](#)
- **Tasks**
 1. Write a five-line pre-registration: hypothesis, model, grid of (d, b) , tests, what would change your mind.
 2. Estimate d of each series and of the spread $\beta' X_t$ by local Whittle, then the FCVAR by profile likelihood with trace tests.
 3. Interpretation: what do the data support, and what would you add to the model?
- **Report:** a five-line plan, five estimates, two tests, a project plan
- **Notebook:** section [C1](#)

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant to summarise long memory and rough volatility. The answer:
- ▣ (a) An ARFIMA with $d = 0.3$ has autocorrelations that decay exponentially, only more slowly than an AR(1).
- ▣ (b) The standard error of local Whittle is $\pi/\sqrt{24m}$.
- ▣ (c) A larger bandwidth m is always better, because the variance falls.
- ▣ (d) If the Qu test rejects, the series has structural breaks.
- ▣ (e) $H = 0.1$ means that volatility has short memory, because $H < 0.5$ is antipersistence.
- ▣ (f) FIGARCH with $0 < d < 1$ is covariance stationary with a finite unconditional variance.
- ▣ (g) Exact local Whittle remains consistent for $d = 1.2$, whereas local Whittle does not.
- ▣ Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
 - ▶ 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- ▣ Long memory makes i.i.d. and HAC standard errors far too small
- ▣ Local Whittle is the default; exact local Whittle when d may exceed $1/2$; always vary m
- ▣ Test against level shifts before claiming long memory
- ▣ Log volatility is rough ($H \approx 0.1$) and persistent; the proxy affects H
- ▣ Forecast gains over HAR are small and horizon-dependent

After the seminar

- ▣ Lecture 10 develops each topic of today: spectral theory, asymptotics of the estimators, the Qu test, FCVAR, FIGARCH, rough volatility and its forecasts
- ▣ Try the [Proposed] tasks in the notebook
- ▣ C1 can grow into a team project: co-memory of Central and Eastern European volatilities
- ▣ Reading: [Robinson \(1995b\)](#); [Shimotsu and Phillips \(2005\)](#); [Qu \(2011\)](#); [Gatheral, Jaisson and Rosenbaum \(2018\)](#)
- ▣ **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

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